Thermal radiation and color

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use peak wavelength as one piece of evidence about a thermal spectrum.

SUCCESS CRITERIA - I can...

1. Relate higher temperature to shorter peak wavelength.
2. Use kelvin in Wien's law.
3. Distinguish spectral peak from perceived color.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "blackbody" and "peak" to explain your model or result. Sentence starters are optional.

blackbody: an ideal absorber/emitter used in a thermal-radiation model

peak: the location of the largest value on a specified graph

kelvin: the absolute temperature scale used in thermal ratios

01 Bring back an earlier idea

An object moves 0.3 m toward a fixed plane mirror. **Find** image movement and the change in object-image separation.

Sentence starter: My design shows … . It meets the requirement by … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $\lambda_{max}\, T = 2.9\times10^{-3}$ m·K.
Ideal blackbody; T in kelvin, λ_{max} in meters. A shorter peak wavelength means higher temperature, not one exact visible color.

Worked reasoning: A thermal spectrum's peak shifts toward shorter wavelength as absolute temperature rises. Wien's relation$\lambda_{peak}\, T \approx 2.9\times10^{-3}$ m·K gives$\lambda_{peak} \approx 9.7$ micrometers at 300 K.

Your turn: Sketch two broad thermal spectra, labeling axes and the hotter peak's shift.

Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: ① ② ③

03 Represent and interpret

Two ideal thermal emitters are at 3000 and 6000 K. Which has the shorter peak wavelength? Does that mean they emit only one color?

Word help: ideal = using stated simplifying conditions

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: ① ② ③

04 Solve and justify

Calculateλ_{max} for 5000 K and 10, 000 K.
Convert to nm.
Explain why you should not assign an exact observed meteor color from the peak alone.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

An object has an infrared peak but looks red-hot.
Explain how both can be true.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

06 Choose a method

Across a stationary boundary, frequency remains 4 Hz while speed changes from 24 to 32 m/s.
Find both wavelengths.

Word help: boundary = the place where a medium or constraint changes

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

07 Exit check

An ideal blackbody has λ_{max}=725 nm.
Estimate temperature.

Word help: estimate = find an approximate value

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Find peak wavelengths at 3000 and 6000 K.
State why wavelength peak alone does not specify one exact perceived color.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3